

Course: EEE F311 Communication Systems
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Date: 19-11-2025

1 Objectives

- SER performance

2 Tutorial Problems

1. Compare the SER performance for 4-ary PAM and QPSK at the same E_b/N_0 at a high E_b/N_0 .
Solution: At high E_b/N_0 (write direct formula for both 4-PAM and QPSK in terms of Q-function)

$$P_e^{(4\text{-PAM})} = \frac{3}{2} Q\left(\sqrt{\frac{4}{5} \frac{E_b}{N_0}}\right)$$

$$P_e^{(\text{QPSK})} \approx 2 Q\left(\sqrt{2 \frac{E_b}{N_0}}\right)$$

Since

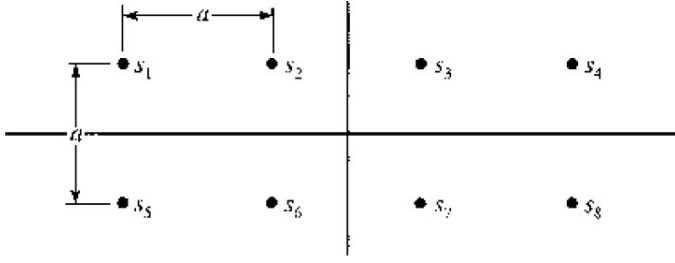
$$\sqrt{2 \frac{E_b}{N_0}} > \sqrt{\frac{4}{5} \frac{E_b}{N_0}}$$

$$\Rightarrow Q\left(\sqrt{2 \frac{E_b}{N_0}}\right) < Q\left(\sqrt{\frac{4}{5} \frac{E_b}{N_0}}\right)$$

$$P_e^{(\text{QPSK})} < P_e^{(4\text{-PAM})}$$

At high $\frac{E_b}{N_0}$, QPSK has lower P_e than 4-PAM.

2. BP Lathi Text book question 10.6-6:
Derive the SER for an 8-QAM constellation under AWGN.



$$\begin{aligned} P(C|m_1) &= P(C|m_4) = P(C|m_5) = P(C|m_8) \\ P(C|m_2) &= P(C|m_3) = P(C|m_6) = P(C|m_7) \end{aligned}$$

$$\begin{aligned} P(C|m_1) &= P\left(n_1 > \frac{a}{2}, n_2 > \frac{-a}{2}\right) \\ &= \left[1 - Q\left(\frac{a}{\sqrt{2N}}\right)\right] \cdot \left[1 - Q\left(\frac{a}{\sqrt{2N}}\right)\right] = \left[1 - Q\left(\frac{a}{\sqrt{2N}}\right)\right]^2 \\ P(C|m_2) &= P\left(|n_1| < \frac{a}{2}, n_2 > \frac{-a}{2}\right) \\ &= \left[1 - 2Q\left(\frac{a}{\sqrt{2N}}\right)\right] \cdot \left[1 - Q\left(\frac{a}{\sqrt{2N}}\right)\right] \end{aligned}$$

$$P(C) = \frac{1}{2} [P(C|m_1) + P(C|m_2)] = \frac{1}{2} \left[1 - Q\left(\frac{a}{\sqrt{2N}}\right)\right] \cdot \left[2 - 3Q\left(\frac{a}{\sqrt{2N}}\right)\right]$$

$$P_{eM} = 1 - P(C) = \frac{1}{2} Q\left(\frac{a}{\sqrt{2N}}\right) \left[5 - 3Q\left(\frac{a}{\sqrt{2N}}\right)\right]$$

In the next page, the representation of a in terms of E_b is provided. If time is short: ask the student to represent a in terms of E_b .

The average symbol power $\overline{E}$ is

$$\begin{aligned}\overline{E} &= \frac{1}{8} [4 (E_{S_1} + E_{S_2})] \\ &= \frac{1}{2} \left[\frac{5a^2}{2} + \frac{a^2}{2} \right] \\ &= \frac{3a^2}{2}\end{aligned}$$

The power per bit E_b is

$$E_b = \frac{\overline{E}}{3} = \frac{a^2}{2}$$

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Therefore

$$a = \sqrt{2E_b}$$

and

$$P_{eM} = \frac{1}{2} Q \left(\sqrt{\frac{E_b}{N}} \right) \left[5 - 3Q \left(\sqrt{\frac{E_b}{N}} \right) \right]$$